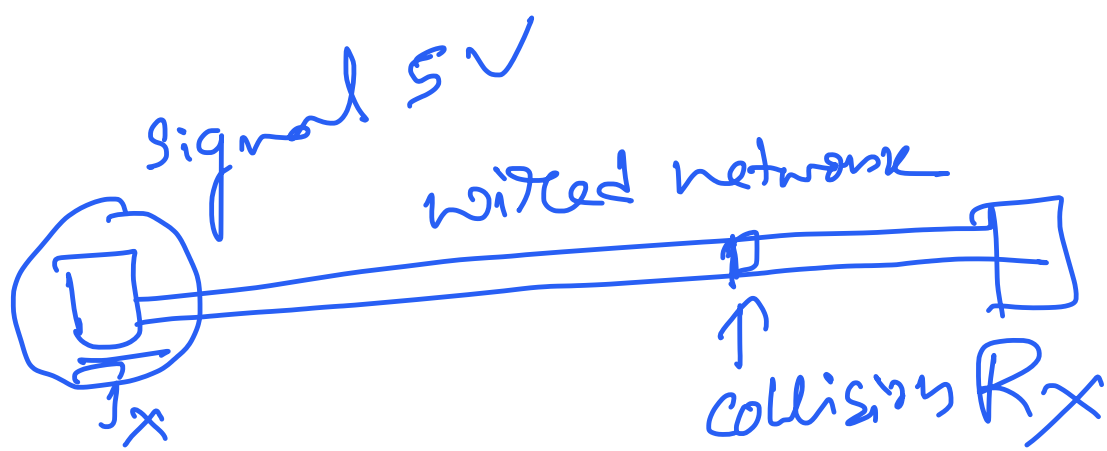
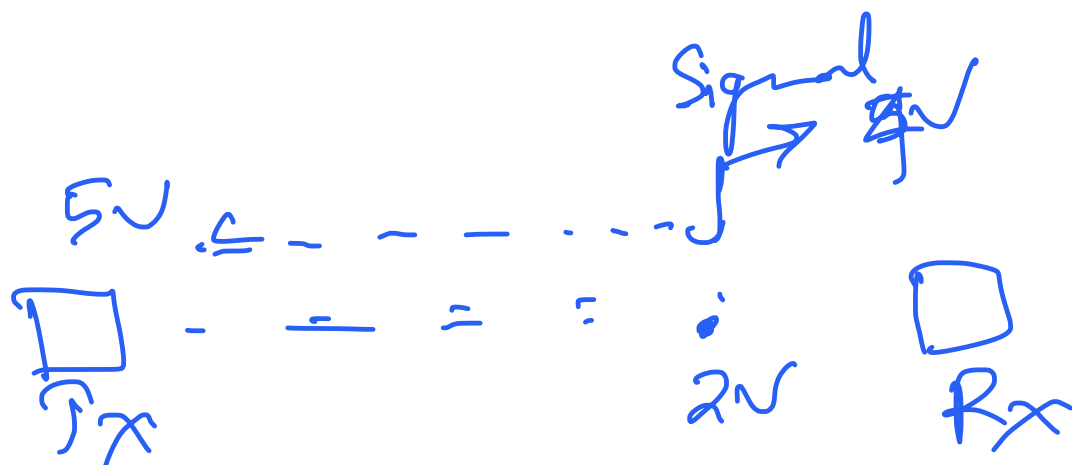




Collision Detection



You can never detect collision in wireless network.

Feedback ACK → Acknowledgment.

CSMA/CA

↳ Carrier Sense Multiple Access/Collision Avoidance

In wired n/w

CSMA/CD

↳ Collision Detect

DIFS  $\rightarrow$  Distributed Interframe Spacing

SIFS  $\rightarrow$  Short " "

Contention window

Backoff.

- Every node listens to the channel for DIFS time.
- If the channel is IDLE
  - $\rightarrow$  It picks a random number between 0 to CW (Contention window)
  - $\rightarrow$  It waits for those many time slots
  - $\rightarrow$  When the backoff counter reaches 0 or expires
  - $\rightarrow$  the node starts transmission.
  - $\rightarrow$  If the packet is received successfully, the receiver sends an ACK after SIFS time

→ If the channel is busy  
Keep on sensing the channel  
till it ~~can~~ becomes free.

Initially  $\omega = \omega_{\min}$   $\omega_{\min} = 16$   
 $\omega_{\max} = 1024$   
 0-16      0-15

→ If the Sender does not receive an ACK after SIFS time, it doubles the CW & picks a random number between 0 - CW

$$C_w = C_{win}$$

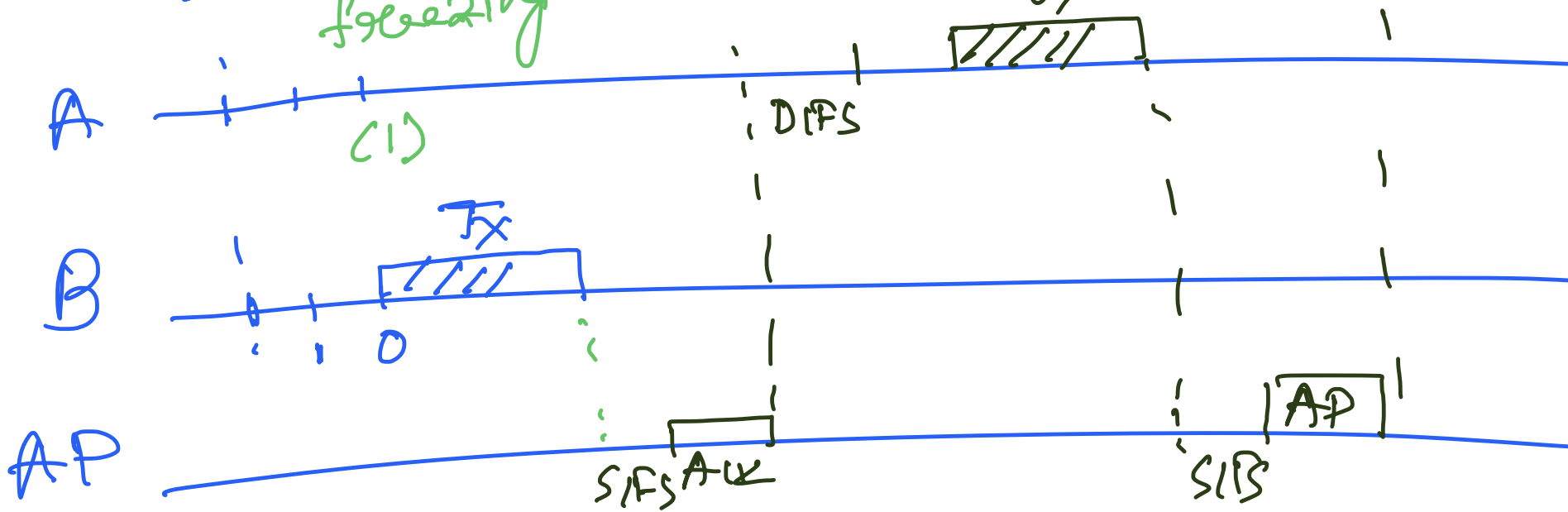
$CW = CW_{min}$   
 for every collision  $CW = 2 \times CW$

|                 |      |
|-----------------|------|
|                 | 16   |
| :               | 32   |
| :               | 64   |
| :               |      |
| $CW = CW_{max}$ | 1024 |

A → AP  
B → AP  
DIFS

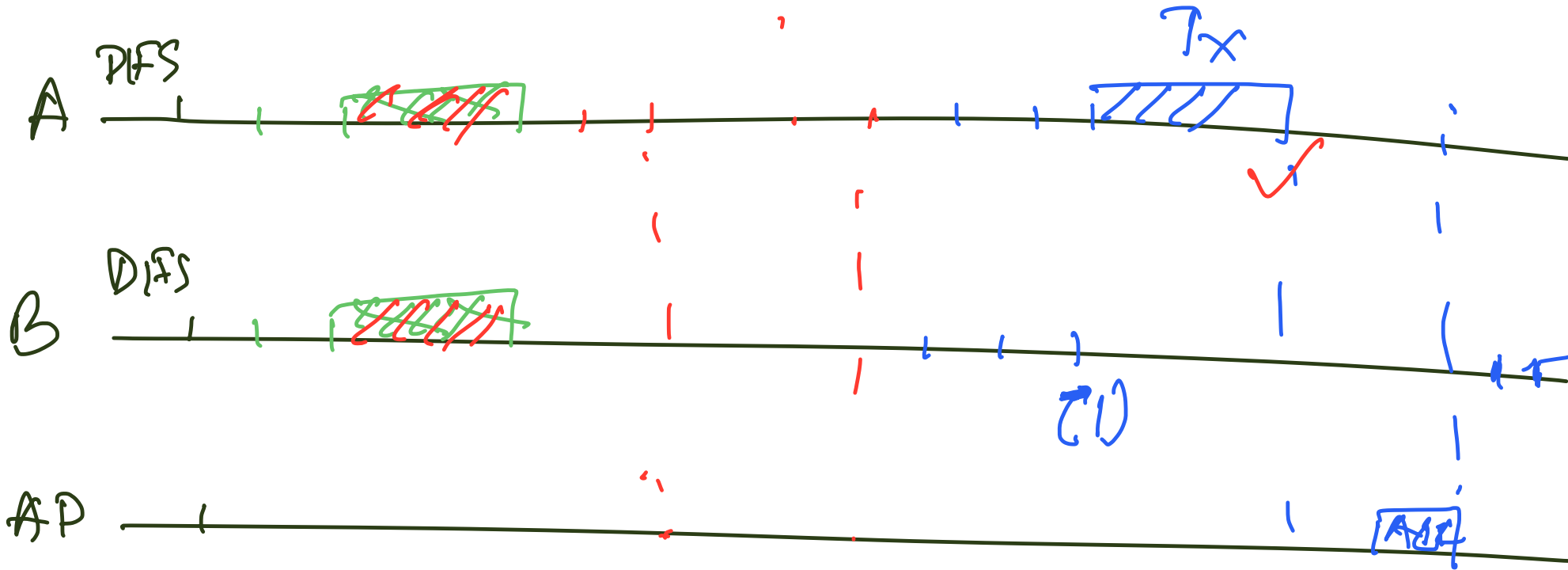
A → 0-15 → 3  
B → 0-15 → 2

freezing of backoff time

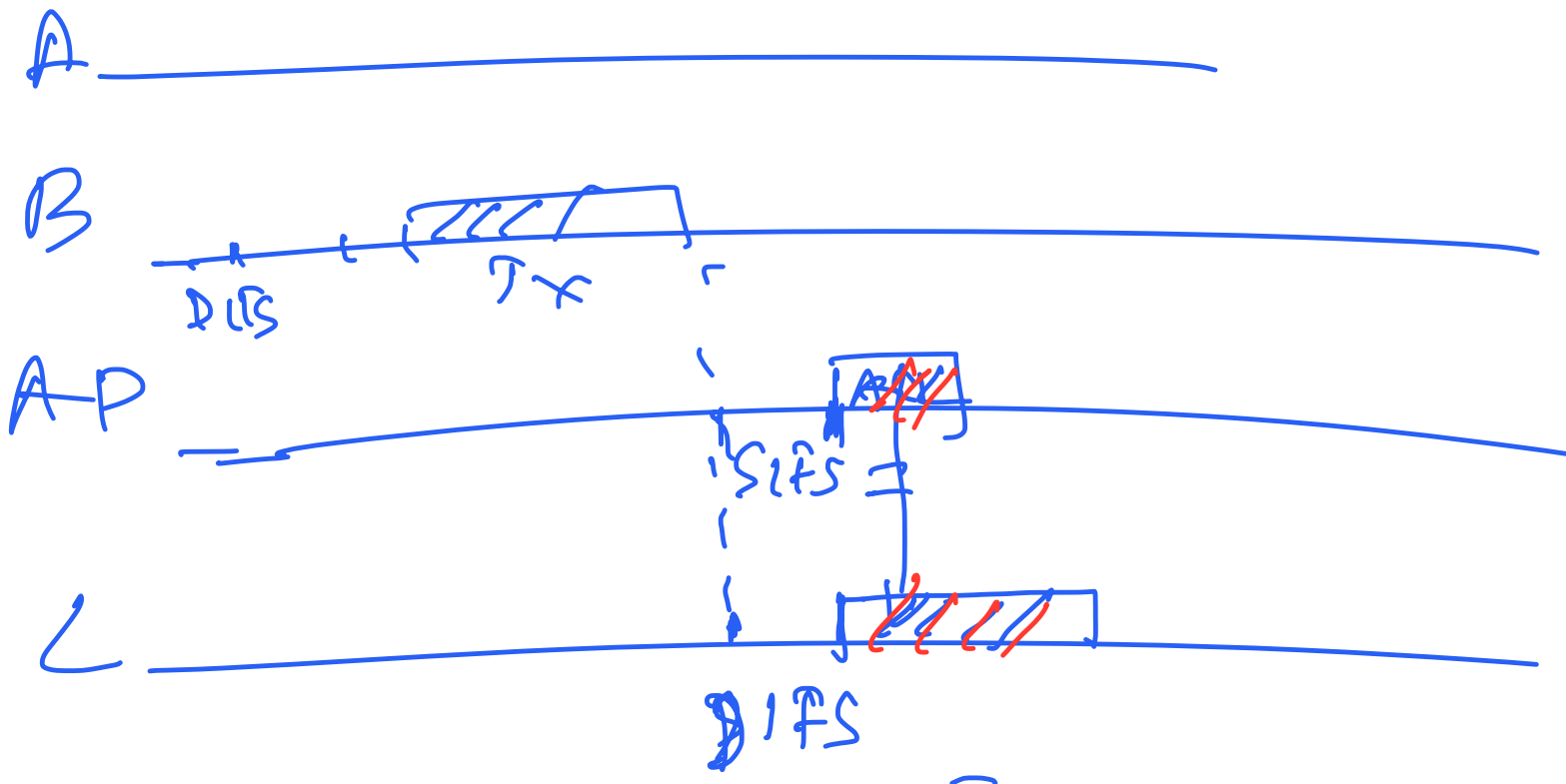


A → 2  
B → 2

CW = 16 A → 0-31 3  
CW = 32 B → 0-31 4  
CW = CW<sub>min</sub>



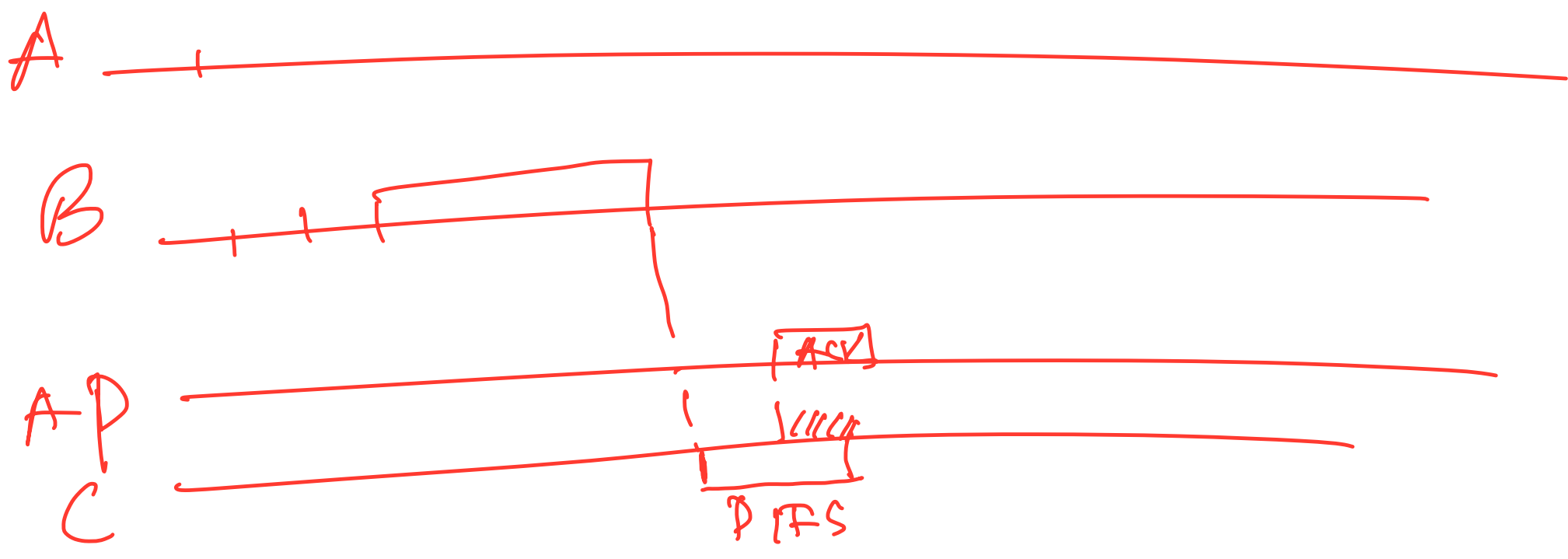
DIFS = SIFS



$$SIFS < DIFS$$

0-CAW

$$DIFS = SIFS + 2 \times \text{Slot time}$$

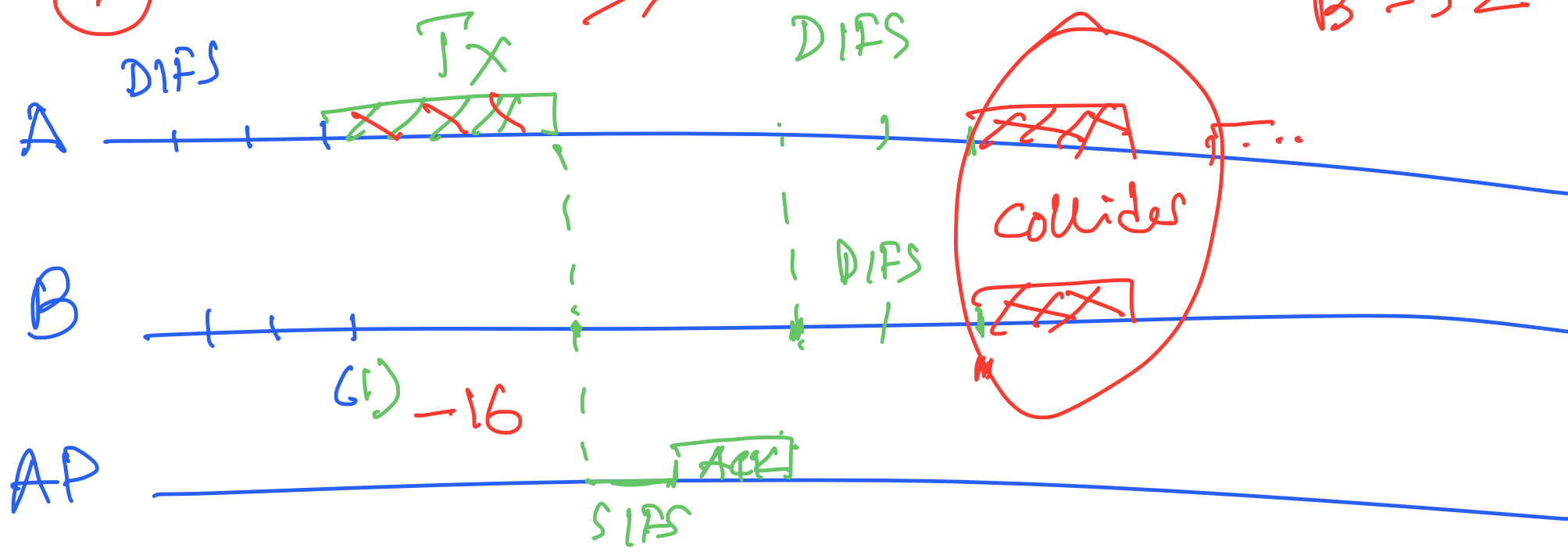


A, B, AP  
↓ ↓ → 1 packet  
2 packet → ③

②  
①

3 2

A = 32  
B = 32



(2) ↓ double 32